

🌀 Section: Particles and Waves

💎 Topic: Wave-Particle Duality – Determining Planck’s Constant

(a) Draw a graph of V against f

✔ 3 marks

Frequency f (10^{14} Hz)	Voltage V (V)
4.5	1.38
5.0	1.62
5.1	1.65
5.3	1.74
6.4	2.32

✔ Graph axes should be:

- X-axis: f , in 10^{14} Hz
- Y-axis: V , in volts
- Points plotted accurately
- Line of best fit (not dot-to-dot)

📌 You’ll need to use the square-ruled paper provided in the exam or graph software.

(b) Calculate the gradient of your graph

✔ 2 marks

Choose two points *on the best-fit line*, for example:

- Point A: $f = 4.5, V = 1.38$
- Point B: $f = 6.4, V = 2.32$

Gradient = $\frac{\Delta V}{\Delta f} = \frac{2.32 - 1.38}{6.4 - 4.5} = \frac{0.94}{1.9} = 0.495 \text{ V}/(10^{14} \text{ Hz})$

(c) Use the gradient to calculate Planck’s constant

✔ 2 marks

From:
 $eV = hf \Rightarrow h = e \cdot \text{gradient}$

$h = 1.60 \times 10^{-19} \cdot 0.495 \times 10^{-14} = 7.92 \times 10^{-34} \text{ Js}$

✔ This is close to the accepted value $6.63 \times 10^{-34} \text{ Js}$
(so the student has done well!)

(d) Suggest one improvement to the experiment

✔ 1 mark

✔ Sample answers:

- Use more LEDs with known frequencies to improve the accuracy of the graph
- Use a voltmeter with smaller uncertainty
- Take repeat readings and average
- Ensure constant temperature to avoid variation in LED characteristics

🧠 Revision Tips

- This experiment is based on the photoelectric effect: $E = hf = eV$
- The gradient of the V vs. f graph gives $\frac{h}{e}$
- Drawing and interpreting straight-line graphs is a critical exam skill!

